

AllSchoolsColleges — Admissions & Growth Analytics Platform

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Project Type: Business Intelligence & Analytics

Technology Stack: Google BigQuery, SQL, Power BI, DAX, Python

1. Purpose

The purpose of this Functional Requirements Document is to translate the business requirements defined in the BRD into specific **functional, data, analytical and reporting requirements** for the AllSchoolsColleges Analytics Platform.

The solution will use **BigQuery for analytical data storage and SQL analysis**, with **Power BI and DAX** providing interactive reporting and business intelligence.

2. Solution Overview

The proposed solution follows the following analytical workflow:

Source Data



Data Preparation



Google BigQuery



SQL Analysis & Aggregation



Power BI Data Model



DAX Measures



Interactive Dashboard



Business Decision Support

The solution will analyze the student journey from:

Lead



Qualified Lead



Application



Enrollment



Student Fee



Commission

3. Functional Requirements

3.1 Lead Management Analytics

FR-001 — Total Lead Calculation

The system shall calculate the total number of leads available within the selected filter context.

Metric:

Total Leads = Count of Lead Records

FR-002 — Lead Source Analysis

The system shall allow users to analyze leads by acquisition source.

Examples include:

- Organic
- PPC / Paid Search

The underlying fact_leads data contains lead source and traffic-related fields.

FR-003 — Lead Quality Analysis

The system shall calculate lead quality based on qualified leads relative to total leads.

Lead Quality Rate =

Qualified Leads / Total Leads

Users shall be able to compare lead quality across acquisition sources.

FR-004 — Lead Trend Analysis

The system shall provide time-based analysis of lead generation.

Users shall be able to analyze:

- Monthly lead trends
- Year-over-year performance
- Lead source trends

4. Application Analytics

FR-005 — Application Volume

The system shall calculate the total number of applications.

FR-006 — Application Status

The system shall allow users to analyze applications by status.

The application data contains statuses including **Applied, Accepted, Rejected and Under Review**.

FR-007 — Application Trend

The system shall allow users to analyze application performance over time.

5. Enrollment Analytics

FR-008 — Total Enrollment

The system shall calculate the total number of enrollments.

FR-009 — Enrollment Status

The system shall distinguish between successful enrollments and withdrawals.

The source contains **Enrolled** and **Withdrawn** enrollment statuses.

FR-010 — Enrollment Rate

The system shall calculate enrollment performance using the appropriate lead/application denominator defined by the dashboard metric.

FR-011 — Withdrawal Rate

The system shall calculate the proportion of enrollment records classified as withdrawn.

FR-012 — Enrollment Trend

The system shall provide monthly and time-based enrollment analysis.

6. PPC & Marketing Analytics**FR-013 — PPC Performance**

The system shall provide PPC performance analysis using:

- Impressions
- Clicks
- Leads
- Applications
- Enrollments
- Spend
- Campaign
- School
- Month

These fields are supported by the fact_ppc_performance source structure.

FR-014 — Campaign Performance

The system shall allow users to compare campaigns based on:

- Leads
 - Applications
 - Enrollments
 - Spend
 - Conversion performance
-

FR-015 — Marketing Spend

The system shall calculate and display total marketing/PPC expenditure.

FR-016 — Cost per Enrollment

The system shall calculate marketing spend relative to successful enrollments.

Cost per Enrollment =
Marketing Spend / Enrollments

FR-017 — Lead Source Performance

The system shall compare Organic and PPC/Paid Search performance using relevant acquisition and conversion metrics.

7. School Analytics

FR-018 — School Performance

The system shall allow users to evaluate individual school performance.

Metrics shall include, where applicable:

- Leads
 - Applications
 - Enrollments
 - Revenue
 - Commission
-

FR-019 — Top School Identification

The system shall identify schools with the highest enrollment performance.

FR-020 — School Comparison

Users shall be able to compare multiple schools using dashboard filters and visualizations.

8. Geographic Analytics

FR-021 — Regional Analysis

The system shall provide enrollment and business performance analysis by region.

FR-022 — State Analysis

The system shall provide state-level enrollment analysis.

FR-023 — Geographic Visualization

The Power BI solution shall provide geographic visualization for relevant school and enrollment performance.

FR-024 — Top Region / State

The system shall identify the region and state generating the strongest enrollment performance.

9. Revenue & Commission Analytics

FR-025 — Student Fee Analysis

The system shall use enrollment fee information for revenue analysis.

The enrollment dataset contains fee_amount associated with enrollment records.

FR-026 — Commission Revenue

The system shall calculate commission revenue using the defined 10% commission business rule.

Commission Revenue =

Applicable Student Fees × 10%

FR-027 — Commission per Enrollment

The system shall calculate average commission generated per enrollment.

Commission per Enrollment =

Commission Revenue / Total Enrollments

FR-028 — Commission ROI

The system shall calculate commission return relative to marketing expenditure.

Commission ROI =

(Commission Revenue - Marketing Spend)

/

Marketing Spend

10. Growth & Targeting Requirements

FR-029 — School Targeting Opportunity

The system shall provide an analytical view to identify schools based on combinations of:

- Lead volume
 - Lead quality
 - Enrollment
 - Commission revenue
-

FR-030 — Regional Growth Opportunity

The system shall allow users to compare regions based on commission opportunity and enrollment performance.

FR-031 — Lead Quality Opportunity

The system shall allow users to identify acquisition sources with relatively stronger or weaker lead quality.

FR-032 — Priority Schools

The system shall provide a school-level view containing relevant performance indicators to support targeting decisions.

11. Dashboard Requirements

The Power BI solution shall contain five analytical pages.

FR-033 — Executive Overview

The dashboard shall provide:

- Total Leads
 - Applications
 - Enrollments
 - Revenue
 - YoY comparisons
 - Overall business trends
-

FR-034 — Admissions Performance

The dashboard shall provide:

- Applications
- Enrollments

- Enrollment rates
 - Free enrollment rate
 - Withdrawal rate
 - Admissions trends
-

FR-035 — Marketing Performance

The dashboard shall provide:

- Marketing spend
 - Lead source analysis
 - Campaign analysis
 - PPC performance
 - Lead quality
 - Marketing efficiency
-

FR-036 — School Performance

The dashboard shall provide:

- School-level enrollment
 - State performance
 - Regional performance
 - Top schools
 - Geographic visualization
-

FR-037 — Growth & Targeting Strategy

The dashboard shall provide:

- Commission revenue
- Commission per enrollment
- Marketing spend
- Marketing spend / commission
- Lead quality rate
- Commission ROI
- School targeting opportunity

- Regional opportunity
 - Priority schools
-

12. Filtering Requirements

FR-038 — Interactive Filters

Users shall be able to filter dashboard information using relevant dimensions such as:

- Date
 - Year
 - Month
 - Region
 - State
 - School
 - Lead Source
 - Campaign
 - Application Status
 - Enrollment Status
-

FR-039 — Cross-Visual Interaction

Dashboard visuals shall respond to selections made in other relevant visuals.

13. Data Requirements

The analytical solution shall utilize the available operational and marketing datasets.

Key fact tables include:

fact_leads

fact_applications

fact_enrollments

fact_ppc_performance

The source confirms that fact_leads includes lead, student, school, date, source, traffic, target, campaign and status information.

The application table contains application, student, school, date, grade and status information.

The enrollment table contains enrollment, student, school, date, status and fee information.

14. BigQuery Requirements

FR-040 — Data Warehouse

The system shall use Google BigQuery as the analytical data warehouse.

FR-041 — SQL Analytics

SQL shall be used to perform:

- Aggregation
 - Trend analysis
 - Funnel analysis
 - Campaign analysis
 - Lead-source analysis
 - School analysis
 - Regional analysis
 - Revenue analysis
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FR-042 — Analytical Queries

The repository shall maintain SQL scripts used for the project's analytical requirements, including:

Master_SQL_Analysis.sql

Campaign_Performance.sql

Lead_Source_Performance.sql

Monthly_Lead_Trend.sql

Monthly_PPC_Trend.sql

Overall_Funnel_Performance.sql

PPC_School_Performance.sql

Revenue_By_Region.sql

School_Performance.sql

school_coverage.sql

15. DAX Requirements

Power BI shall use DAX measures for business calculations that require dynamic filter context.

Examples include:

Commission Revenue =

[Total Revenue] * 0.10

Commission per Enrollment =

DIVIDE(

[Commission Revenue],

[Total Enrollments]

)

Lead Quality Rate =

DIVIDE(

[Qualified Leads],

[Total Leads]

)

Commission ROI =

DIVIDE(

[Commission Revenue] - [Total Marketing Spend],

[Total Marketing Spend]

)

16. Data Quality Requirements

FR-043 — Duplicate Records

The analytical pipeline should identify potential duplicate business records where applicable.

FR-044 — Missing Values

Missing values shall be identified and handled according to the business meaning of the affected field.

FR-045 — Referential Integrity

Foreign-key relationships such as:

student_id

school_id

campaign_id

should be validated where applicable.

FR-046 — Date Validation

Date fields shall be validated to ensure they can support time-based analysis.

17. Validation & Testing

The following validation activities shall be performed:

Test	Objective
Record Count Validation	Confirm expected data volume
Duplicate Check	Identify duplicate records
Null Check	Identify missing critical fields
Relationship Validation	Validate table relationships
KPI Reconciliation	Compare dashboard metrics with source/SQL results
DAX Validation	Verify calculated measures
Filter Testing	Ensure filters affect relevant visuals
Visual Validation	Confirm correct dimensions and measures
Business Rule Testing	Validate 10% commission calculation

18. Acceptance Criteria

The solution will be considered functionally complete when:

- All five Power BI pages are available.
 - Required KPIs are calculated correctly.
 - Lead, application and enrollment analysis is available.
 - Marketing and PPC analysis is available.
 - School and geographic analysis is available.
 - Commission calculations follow the 10% business rule.
 - Users can filter the dashboard.
 - Visual interactions work correctly.
 - SQL analytical queries produce expected results.
 - Dashboard metrics reconcile with the underlying data.
 - Growth and targeting analysis is available.
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19. Expected User Workflow

Executive Overview



Understand Overall Performance



Admissions Performance



Identify Funnel Issues



Marketing Performance



Evaluate Acquisition Efficiency



School Performance



Identify Strong Schools / Regions



Growth & Targeting Strategy



Determine Where to Focus Next

20. Traceability

The functional solution maps directly back to the business requirements:

Business Requirement	Functional Solution
Monitor business performance	Executive Overview
Analyze admissions	Admissions Performance
Evaluate marketing	Marketing Performance
Analyze schools	School Performance
Analyze geography	State/Region visuals + map
Measure financial opportunity	Commission metrics

Business Requirement	Functional Solution
Improve lead quality	Lead Quality analysis
Identify growth opportunities	Growth & Targeting Strategy

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